

TECHBIZ SECURITY

Cyber Security Internship Program

WEEK 2 – TASK 4 (SOC)

Introduction to Log Analysis and Linux Security Logs

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1. Executive Summary

This report documents the practical work completed during Week 2, Task 4 of the Cyber Security Internship at TechBiz Security. The task focused on developing a foundational understanding of Linux logging, identifying common log locations on a Linux system, reviewing system activity using journalctl, examining authentication and privileged-access events, and performing basic keyword searches across logs using command-line utilities such as grep and find.

Log analysis is a core function within a Security Operations Center (SOC), as logs provide the raw evidence needed for monitoring, investigation, and incident response. This task provided hands-on exposure to the log sources and command-line techniques that form the starting point for these SOC activities, supported by structured learning through TryHackMe's Linux Fundamentals and Intro to Logs rooms.

2. Objectives

- Understand the purpose of security and system logs.
- Identify common Linux log locations.
- Explore the /var/log directory.
- Understand different types of Linux log files.
- Use journalctl to review system logs.
- Examine authentication-related events.
- Use grep to search logs for relevant keywords.
- Understand how logs support SOC monitoring and investigation.
- Identify the importance of centralized and properly retained logs.

3. Introduction to Logging

A log is a time-stamped record generated by an operating system, application, or service to document an event that occurred on that system. Organizations generate logs to maintain visibility into system behaviour, user activity, and service operations, and to preserve a historical record that can be reviewed when something needs to be verified or investigated.

Logs capture a wide range of activity, including system boot and shutdown events, application errors, authentication attempts, administrative commands, and service status changes. For a SOC analyst, logs are one of the most important sources of evidence available, since they allow an analyst to:

- Detect suspicious or anomalous activity.
- Investigate security incidents after they are reported or flagged.
- Identify authentication attempts, including failed logins.
- Track administrative actions such as sudo usage.
- Understand normal versus abnormal system behaviour.
- Establish a timeline of events during an investigation.

4. Linux Logging Environment

The practical exercise was carried out on a Kali Linux system. Most Linux distributions, including Kali, centralise the majority of system and application log files under a single directory: /var/log. This directory typically contains logs generated by the kernel, system services, package management, authentication mechanisms, and various installed applications.

The first stage of the practical activity involved navigating to /var/log and reviewing its contents. From a SOC perspective, becoming familiar with this directory is an important first step, since it establishes where relevant evidence can be found before any deeper investigation or log search is carried out.

5. Practical Activity 1 – Exploring /var/log

The command `ls -lah` was used inside `/var/log` to produce a detailed, human-readable directory listing. This command lists all files, including hidden ones, along with their permissions, ownership, size, and last-modified timestamp.

The listing shown in the evidence includes a range of log files and log-related directories, such as `alternatives.log`, `boot.log`, `dpkg.log`, `fontconfig.log`, `macchanger.log`, `lastlog`, `wtmp`, `btmpt`, `Xorg.0.log`, and the journal directory used by `systemd`. It also shows several application- and service-specific subdirectories, including `apache2`, `mysql`, `nginx`, `openvpn`, and `postgresql`, along with directories associated with authentication-related logging. Not every listed file was analysed in detail during this task; the objective at this stage was to identify what log sources exist on the system.

Reviewing `/var/log` in this way is important for a SOC analyst because it provides a quick overview of the log sources available on a host, along with useful metadata such as file size and last-modified time, which can indicate how active a particular log source is.

```
manuaria@kali: /var/log
$ ls -lah
total 9.0M
drwxr-xr-x 22 root root 4.0K Aug 27 18:30 .
drwxr-xr-x 12 root root 4.0K Jun 15 2025 ..
-rw-r--r-- 1 root root 63K Aug 4 11:16 alternatives.log
-rw-r--r-- 1 root root 215K Dec 14 2025 alternatives.log.1
drwxr-xr-x 2 root adm 4.0K May 25 18:54 apache2
drwxr-xr-x 2 root root 4.0K Aug 4 11:16 apt
drwxr-xr-x 27 10:30 boot.log
drwxr-xr-x 2 10:54 boot.log.1
-rw-r--r-- 1 root root 15K Mar 13 11:55 btmap.1
drwxr-xr-x 1 root root 3.5M Feb 19 2026 dpkg.log.1
drwxr-xr-x 1 root root 8.3K Aug 4 11:13 fontconfig.log
drwxr-xr-x 2 gsm gsm 4.0K May 20 2025 gsm
drwxr-xr-x 3 inetisim inetisim 4.0K Jun 15 2025 inetisim
drwxr-xr-x 3 root root 4.0K Jun 15 2025 installer
drwxr-xr-x 3 root root 4.0K Jun 15 2025 journal
-rw-r--r-- 1 root root 0 Jun 15 2025 lastlog
drwxr-xr-x 27 10:31 lightdm
drwxr-xr-x 1 root root 0.1K Aug 27 18:30 macchanger.log
-rw-r--r-- 1 root root 157 May 25 18:54 macchanger.log.1.gz
drwxr-xr-x 2 mosquito mosquito 4.0K Mar 21 2025 mosquito
drwxr-xr-x 2 mysql mysql 4.0K Aug 1 20:15 mysql
drwxr-xr-x 2 root adm 4.0K Jun 15 2025 nginx
drwxr-xr-x 2 root root 4.0K Aug 1 20:17 nmap-scanner
drwxr-xr-x 2 root root 4.0K Apr 3 2025 openvpn
drwxr-xr-x 2 root root 4.0K May 25 10:54 private
drwxr-xr-x 2 root root 4.0K Jun 15 2025 private
-rw-r--r-- 1 root root 39 Jun 15 2025 README -> ../usr/share/doc/systemd/README.logs
drwxr-xr-x 2 redis redis 4.0K Jun 15 2025 redis
drwxr-xr-x 4 root root 4.0K Sep 20 2025 runit
drwxr-xr-x 2 root adm 4.0K Apr 18 2025 samba
drwxr-xr-x 2 speech-dispatcher speech-dispatcher 4.0K Aug 1 20:17 speech-dispatcher
drwxr-xr-x 2 stunnel stunnel 4.0K Jun 15 2025 stunnel
drwxr-xr-x 2 root root 4.0K Nov 18 2025 sysstat
-rw-r--r-- 1 root root 227K Aug 27 18:31 wtmp
-rw-r--r-- 1 root root 26K Aug 27 18:36 Xorg.0.log
-rw-r--r-- 1 root root 26K Aug 26 12:37 Xorg.0.log.old
```

Figure 1: Linux `/var/log` directory listing

6. Practical Activity 2 – Searching for Log Files

Following the directory listing, the `find` command was used with the syntax `find /var/log -maxdepth 1 -type f -printf '%f\n'` to search specifically for regular log files located directly within `/var/log`, excluding subdirectories. This produced a filtered list of file names such as `alternatives.log.1`, `dpkg.log`, `macchanger.log`, `Xorg.0.log`, `wtmp`, `btmpt`, and `boot.log`.

This command demonstrates how an analyst can programmatically enumerate available log files rather than relying solely on a manual directory listing. Being able to quickly and reliably identify available log sources is an important first step during log analysis, particularly when working across multiple hosts or when preparing to search several logs for a specific indicator.

```

manuaria@kali: /var/log
Session Actions Edit View Help

-rw-r--r-- 1 root utmp 1.9K Aug 17 15:46 btmp
-rw-r--r-- 1 root utmp 15K Mar 13 11:55 btmp.1
-rw-r--r-- 1 root rsyslog 4.0K Aug 4 11:16 dpkg.log
-rw-r--r-- 1 root rsyslog 3.5M Feb 19 2026 dpkg.log.1
-rw-r--r-- 1 root rsyslog 8.3K Aug 4 11:13 fontconfig.log
drwxr-xr-x 2 gdm gdm 4.0K May 20 2025 gdm
drwxr-xr-x 3 inetsim inetsim 4.0K Jun 15 2025 inetsim
drwxr-xr-x 3 root root 4.0K Jun 15 2025 installer
drwxr-xr-x 3 root root system-journal 4.0K Jun 15 2025 journal
-rw-r--r-- 1 root utmp 0 Jun 15 2025 lastlog
drwxr-xr-x 2 root root 4.0K Aug 27 18:11 lightdm
-rw-r--r-- 1 root rsyslog 0.1K Aug 27 18:38 macchanger.log
-rw-r--r-- 1 root rsyslog 157 May 25 18:43 macchanger.log.1.gz
drwxr-xr-x 2 macquitto macquitto 4.0K Aug 1 20:11 mysql
drwxr-xr-x 2 root root 4.0K Jun 15 2025 nginx
drwxr-xr-x 2 gdm gdm 4.0K Aug 1 20:11 netatm-scanner
drwxr-xr-x 2 root root 4.0K Apr 3 2025 openvpn
drwxr-xr-x 2 root postgres 4.0K May 25 18:51 private
drwxr-xr-x 2 root root 4.0K Jun 15 2025 README
lrwxrwxrwx 1 root root 39 Jun 15 2025 README -> ../usr/share/doc/systemd/README.logs
drwxr-xr-x 2 root root 4.0K Jun 15 2025 rsyslog
drwxr-xr-x 4 root root 4.0K Sep 20 2025 runit
drwxr-xr-x 2 root root 4.0K Apr 18 2025 samba
drwxr-xr-x 2 speech-dispatcher root 4.0K Aug 1 20:11 speech-dispatcher
drwxr-xr-x 2 stunnel stunnel 4.0K Jun 15 2025 stunnel
drwxr-xr-x 2 root root 4.0K Nov 18 2025 sysstat
-rw-r--r-- 1 root utmp 327K Aug 27 18:31 wtmp
-rw-r--r-- 1 root rsyslog 26K Aug 27 18:38 xorg.0.log
-rw-r--r-- 1 root rsyslog 28K Aug 26 12:19 xorg.0.log.old

[manuaria@kali]~/var/log
└─$ find /var/log -maxdepth 1 -type f -printf '%f\n'
alternatives.log.1
dpkg.log
macchanger.log
fontconfig.log
xorg.0.log.old
btmp
dpkg.log.1
xorg.0.log
wtmp
boot.log
btmp.1
macchanger.log.1.gz
boot.log.1
alternatives.log
lastlog

[manuaria@kali]~/var/log

```

Figure 2: Searching for available log files under /var/log

7. Practical Activity 3 – System Log Analysis Using journalctl

journalctl is the command-line utility used to query and display logs collected by the systemd journal. In the evidence, the command `sudo journalctl -n 20` was used to display the most recent 20 entries recorded in the journal.

The output includes structured information for each log entry, including the timestamp, hostname (kali), the process or service name generating the event (for example, sudo, CRON, kernel, and systemd), and a description of the event itself. Several entries relate to sudo session activity, cron-triggered PHP session cleanup, and kernel-level clipboard handling within the virtual machine. journalctl allows a SOC analyst to review this type of system and service activity centrally, without needing to open multiple individual log files.

For SOC monitoring purposes, the key fields to pay attention to in journal output are the timestamp, host information, process or service name, the user associated with the event, the command executed, and whether the event represents a success or failure. No security incident is indicated by the entries captured in this evidence; the activity shown reflects normal system and administrative operation.

```

[manuaria@kali]~/var/log
└─$ sudo journalctl -n 20
[sudo] password for manuaria:
Aug 27 18:37:35 kali sudo[3854]: pam_unix(sudo:session): session opened for user root(uid=0) by manuaria(uid=1000)
Aug 27 18:38:31 kali sudo[3869]: manuaria : TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl -n 20
Aug 27 18:38:31 kali sudo[3869]: pam_unix(sudo:session): session opened for user root(uid=0) by manuaria(uid=1000)
Aug 27 18:39:01 kali CRON[3879]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Aug 27 18:39:01 kali CRON[3879]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi)
Aug 27 18:39:03 kali kernel: 05:09:03.478133 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (id:FmtX11=0, fmtX11=0, atom=0)
Aug 27 18:39:03 kali kernel: 05:09:03.479067 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (id:FmtX11=0, fmtX11=0, atom=0)
Aug 27 18:39:04 kali kernel: 05:09:04.483493 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (id:FmtX11=0, fmtX11=0, atom=0)
Aug 27 18:39:04 kali kernel: 05:09:04.493666 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (id:FmtX11=0, fmtX11=0, atom=0)
Aug 27 18:39:04 kali kernel: 05:09:04.209310 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (id:FmtX11=0, fmtX11=0, atom=0)
Aug 27 18:39:04 kali systemd[1]: phpsessionclean.service: Deactivated successfully.
Aug 27 18:39:04 kali systemd[1]: Finished phpsessionclean.service - Clean php session files.
Aug 27 18:39:25 kali kernel: 05:09:04.228661 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (id:FmtX11=0, fmtX11=0, atom=0)
Aug 27 18:39:25 kali kernel: 05:09:25.754146 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (id:FmtX11=0, fmtX11=0, atom=0)
Aug 27 18:39:25 kali kernel: 05:09:25.754146 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (id:FmtX11=0, fmtX11=0, atom=0)
Aug 27 18:40:07 kali sudo[3976]: manuaria : TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl -n 20
Aug 27 18:40:07 kali sudo[3976]: pam_unix(sudo:session): session opened for user root(uid=0) by manuaria(uid=1000)
Aug 27 18:40:15 kali sudo[3987]: pam_unix(sudo:session): session opened for user root(uid=0) by manuaria(uid=1000)

[manuaria@kali]~/var/log
└─$

```

Figure 3: Reviewing Linux system logs using journalctl

8. Practical Activity 4 – Authentication and Privileged Activity Analysis

Authentication-related log entries provide visibility into user login activity and privileged operations on a Linux system. In the evidence, journalctl was filtered using `_COMM=sudo`, and further filtered with `grep -Ei "pam_unix|authentication|sudo"` to isolate authentication and sudo-related events from the wider journal output.

The filtered output shows repeated sudo session activity for the user manuarya, including the specific command executed (for example, `/usr/bin/journalctl`, `/usr/bin/less auth.log`, and `/usr/bin/systemctl status ssh`), the terminal (TTY) used, the working directory, and pam_unix session-open and session-close events confirming that each privileged session was properly opened and closed under root.

Monitoring this type of activity is an important SOC function, since it provides visibility into:

- Successful authentication events.
- Failed authentication attempts.
- sudo usage and the specific commands executed with elevated privileges.
- The user accounts involved in privileged operations.
- Session creation and termination for administrative activity.

Consistent monitoring of authentication and sudo logs allows an analyst to establish a baseline of normal administrative behaviour, which makes it easier to identify unusual privilege escalation or unauthorized administrative activity if it occurs.

```
manuarya@kali: /var/log
manuarya@kali:~$ sudo journalctl | grep -Ei "pam_unix|authentication|sudo" | tail -30
Aug 27 10:37:26 kali sudo[3845]: pam_unix(sudo:session): session closed for user root
Aug 27 10:37:35 kali sudo[3854]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/less auth.log
Aug 27 10:37:35 kali sudo[3854]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:37:35 kali sudo[3854]: pam_unix(sudo:session): session closed for user root
Aug 27 10:38:31 kali sudo[3869]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl -n 20
Aug 27 10:38:31 kali sudo[3869]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:40:07 kali sudo[3978]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl -n 20
Aug 27 10:40:07 kali sudo[3978]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:40:07 kali sudo[3978]: pam_unix(sudo:session): session closed for user root
Aug 27 10:40:15 kali sudo[3987]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl -n 20
Aug 27 10:40:15 kali sudo[3987]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:40:15 kali sudo[3987]: pam_unix(sudo:session): session closed for user root
Aug 27 10:47:08 kali sudo[4069]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl
Aug 27 10:47:08 kali sudo[4069]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:47:10 kali sudo[4069]: pam_unix(sudo:session): session closed for user root
Aug 27 10:47:26 kali sudo[4076]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl
Aug 27 10:47:26 kali sudo[4076]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:47:27 kali sudo[4076]: pam_unix(sudo:session): session closed for user root
Aug 27 10:47:40 kali sudo[4084]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl
Aug 27 10:47:40 kali sudo[4084]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:47:42 kali sudo[4084]: pam_unix(sudo:session): session closed for user root
Aug 27 10:48:05 kali sudo[4097]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl
Aug 27 10:48:05 kali sudo[4097]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:48:06 kali sudo[4097]: pam_unix(sudo:session): session closed for user root
Aug 27 10:49:58 kali sudo[4126]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/systemctl status ssh
Aug 27 10:49:58 kali sudo[4126]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:49:58 kali sudo[4126]: pam_unix(sudo:session): session closed for user root
Aug 27 10:50:39 kali sudo[4139]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl _COMM=sudo
Aug 27 10:50:39 kali sudo[4139]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)

manuarya@kali:~$ sudo journalctl | grep "pam_unix(authentication:sudo)" | tail -30
Aug 27 10:39:01 kali CRON[3879]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Aug 27 10:39:01 kali CRON[3879]: pam_unix(cron:session): session closed for user root
Aug 27 10:40:07 kali sudo[3978]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl -n 20
Aug 27 10:40:07 kali sudo[3978]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:40:07 kali sudo[3978]: pam_unix(sudo:session): session closed for user root
Aug 27 10:40:15 kali sudo[3987]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl -n 20
Aug 27 10:40:15 kali sudo[3987]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:40:15 kali sudo[3987]: pam_unix(sudo:session): session closed for user root
Aug 27 10:45:01 kali CRON[4050]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Aug 27 10:45:01 kali CRON[4050]: pam_unix(cron:session): session closed for user root
Aug 27 10:47:08 kali sudo[4069]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl
Aug 27 10:47:08 kali sudo[4069]: pam_unix(sudo:session): session opened for user root(uid=0) by manuarya(uid=1000)
Aug 27 10:47:10 kali sudo[4069]: pam_unix(sudo:session): session closed for user root
Aug 27 10:47:26 kali sudo[4076]: manuarya: TTY=pts/2 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/journalctl
```

Figure 4: Authentication and privileged activity log analysis

9. Practical Activity 5 – Searching for Failed or Relevant Events

The `grep` utility was used to search the systemd journal for a specific keyword using the command `sudo journalctl | grep 'Failed'`. `grep` is a pattern-matching tool that allows an analyst to quickly locate all lines within a log source that contain a given keyword or pattern, without having to manually read through the entire log.

Searching for the term "Failed" returned a range of entries, including failures to read SMBIOS data, failures to retrieve battery percentage from UPower, failures to write to the `utmpx` file, and failed systemd service starts such as `NetworkManager-wait-online.service`. This demonstrates the log-searching technique rather than a specific security event; the entries returned relate to routine service and hardware-interaction failures rather than an authentication compromise.

In a SOC context, searching logs for keywords such as "Failed" is a routine analytical technique. When applied specifically to authentication logs, repeated failed-login entries can indicate incorrect credentials, user error, brute-force attempts, password spraying, or unauthorized access attempts. This evidence demonstrates the search technique that would be applied in that scenario, rather than proving that an actual attack occurred on this system.



These fundamentals directly supported the practical log-analysis activity carried out in this task, since navigating to `/var/log`, listing directory contents, and running commands such as `find`, `journalctl`, and `grep` all rely on comfortable use of the Linux command line. Linux navigation, file handling, and shell usage are foundational skills for any SOC analyst working with Linux-based systems and log sources.



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This exercise reinforced the same grep-based searching technique applied later in the log-analysis activities described in this report. Practical command-line skills of this kind, including directory navigation, file inspection, and keyword searching, directly support security monitoring and investigation tasks carried out by a SOC analyst.

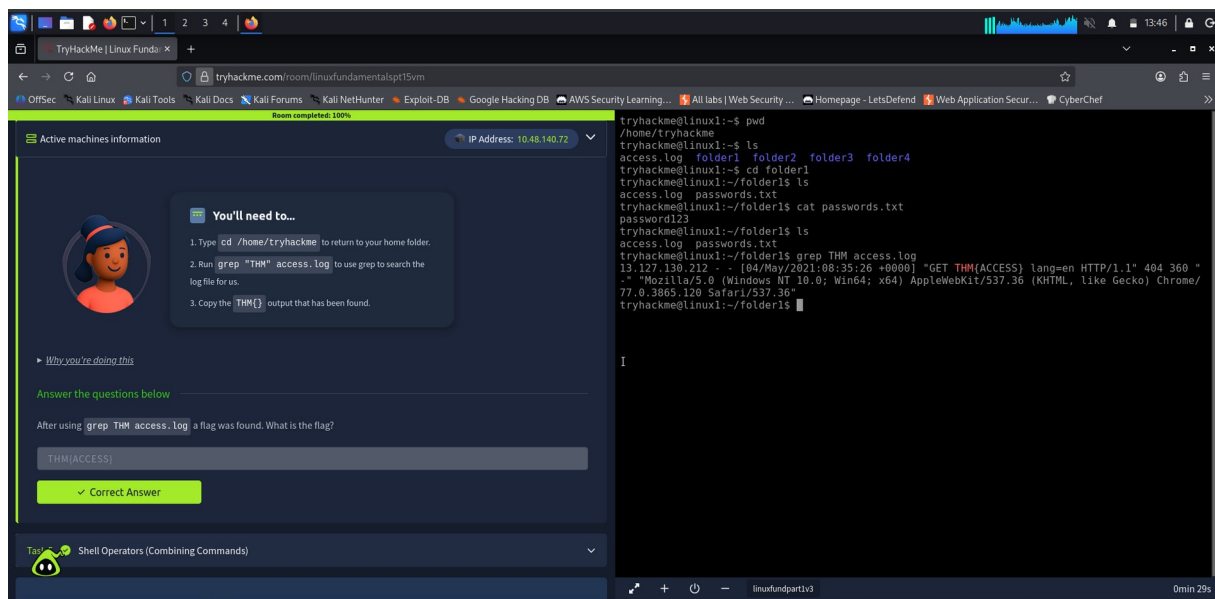


Figure 7: TryHackMe Linux Fundamentals practical task evidence

12. TryHackMe Intro to Logs – Completion

The TryHackMe "Intro to Logs" room was completed to 100%, as shown in the evidence. This room covers the conceptual foundations of logging that support the practical activities carried out in this task, including:

- Introduction to logging and its role in IT and security operations.
- Logs as evidence of historical activity on a system.
- Log types, formats, and standards.
- Log collection, management, and centralisation.
- Log storage, retention, and deletion.

These concepts are directly relevant to SOC operations, as they explain why logs are collected in the first place, how they should be managed and centralised across an environment, and how long they should be retained to support both operational monitoring and forensic investigation.

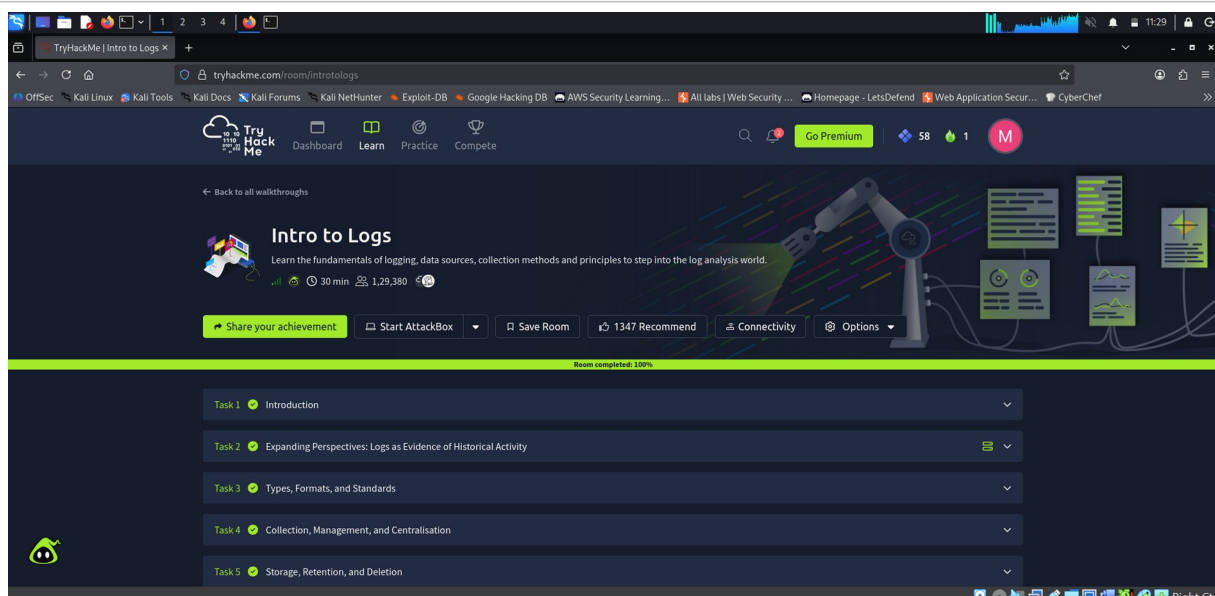


Figure 8: TryHackMe Intro to Logs room completion

13. SOC Relevance of Log Analysis

Log analysis is embedded throughout daily SOC operations. The practical exercises completed in this task map to the early stages of a broader SOC log-analysis workflow, which typically follows this sequence:

- Log Collection – gathering log data from hosts, applications, and services.
- Log Storage – retaining logs in a location where they can be securely accessed.
- Log Monitoring – reviewing logs on an ongoing basis for anomalies.
- Event Searching – using tools such as grep to locate specific events or keywords.
- Event Correlation – relating events across multiple log sources or hosts.
- Suspicious Activity Identification – flagging events that deviate from expected behaviour.
- Investigation – examining flagged events in greater depth.
- Incident Response – acting on confirmed security incidents.
- Documentation – recording findings, actions taken, and outcomes.

A SOC analyst uses authentication logs, system logs, application logs, and, in more mature environments, centralized logging platforms (such as a SIEM) to investigate suspicious behaviour. The Linux command-line exercises completed in this task — exploring `/var/log`, using `journalctl`, and applying `grep` for keyword searching — represent entry-level SOC responsibilities that form the basis for more advanced monitoring and investigation work.

14. Important Log Analysis Commands

Command	Purpose	SOC Relevance
<code>ls -lah /var/log</code>	Lists all files in <code>/var/log</code> with permissions, ownership, size, and modification time.	Provides an initial overview of available log sources on a host.
<code>find /var/log -maxdepth 1 -type f</code>	Enumerates regular log files directly within a directory.	Quickly identifies available log files for further analysis.
<code>sudo journalctl -n 20</code>	Displays the most recent entries in the <code>systemd</code> journal.	Allows review of recent system and service activity.
<code>sudo journalctl _COMM=sudo</code>	Filters journal entries generated by the <code>sudo</code> command.	Isolates privileged-command activity for review.
<code>grep -Ei "pattern"</code>	Searches input for lines matching a given pattern, case-insensitively.	Locates specific keywords, such as authentication or failure events, within logs.
<code>sudo journalctl grep 'Failed'</code>	Searches the full journal output for the keyword "Failed".	Identifies failure events that may warrant further investigation.

15. Key Observations

- Linux stores many system and application logs under `/var/log`.
- Different log sources provide different types of information, from package management to authentication and service activity.
- `journalctl` provides centralized access to `systemd` journal events.
- Authentication and `sudo` activity, including session open and close events, are valuable for security monitoring.
- `grep` can quickly locate relevant keywords within large volumes of log data.
- Log timestamps help establish a timeline of events, which is important during investigations.
- Linux command-line skills are important for SOC analysts working with Linux-based systems.
- Proper log collection, storage, retention, and analysis are essential for effective security monitoring.

16. Security Importance

Reliable logging is a foundational requirement for effective security operations. Organizations depend on properly maintained logs to support several critical functions, including:

- Threat detection – identifying indicators of malicious or unauthorized activity.
- Incident investigation – reconstructing what occurred during a security event.
- Accountability – attributing actions to specific users or processes.
- Forensic analysis – preserving evidence for detailed post-incident review.
- Compliance – meeting regulatory and organizational logging requirements.
- Timeline reconstruction – establishing the sequence of events during an investigation.
- Detection of unauthorized access – identifying login attempts or access that should not have occurred.
- Monitoring privileged activity – tracking the use of administrative or elevated commands.

Without reliable and properly retained logs, a SOC would have limited ability to detect, investigate, or respond to security incidents in a timely and defensible manner.

17. Challenges and Learning Outcomes

This task provided practical, hands-on exposure to core Linux logging concepts and command-line log analysis techniques. Key learning outcomes from the task include:

- Improved understanding of common Linux log locations, particularly `/var/log`.
- Improved command-line log searching using `find` and `grep`.
- A better understanding of authentication and sudo-related events.
- Practical understanding of `journalctl` and the `systemd` journal.
- A clearer understanding of how logs support security investigations.
- Improved SOC-oriented analytical thinking when reviewing log output.

This task represents an introductory level of log analysis; it did not involve advanced correlation, SIEM tooling, or incident response, and no claim of advanced expertise is made.

18. Conclusion

This task provided practical exposure to Linux logging and basic log analysis techniques, including exploring the `/var/log` directory, using `journalctl` to review system activity, examining authentication and sudo-related events, and applying `grep` to search logs for relevant keywords.

Understanding log sources, authentication activity, system events, and command-line searching forms an important foundation for SOC monitoring and incident investigation. The structured learning completed through TryHackMe's Linux Fundamentals and Intro to Logs rooms reinforced the command-line skills and logging concepts applied throughout this task.

19. Evidence Summary

Figure	Evidence	Purpose
Figure 1	<code>/var/log</code> directory listing	Identifying Linux log sources
Figure 2	Log file search	Locating available log files
Figure 3	<code>journalctl</code> output	Reviewing system events
Figure 4	Authentication log analysis	Reviewing authentication and sudo activity
Figure 5	Failed event search	Searching for relevant security events
Figure 6	Linux Fundamentals completion	Supporting Linux command-line skills
Figure 7	Linux Fundamentals practical evidence	Demonstrating command-line practice

Figure	Evidence	Purpose
Figure 8	Intro to Logs completion	Demonstrating completion of log-analysis learning

20. Final Professional Summary

This task delivered practical, entry-level exposure to Linux log locations, system log review, authentication monitoring, and command-line log searching. These skills form a necessary foundation for a Cyber Security Intern working toward SOC responsibilities, where the ability to locate, read, and search logs efficiently is a prerequisite for effective monitoring, investigation, and incident response.